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Comments:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lee, Bang Ning (U TX, M.D. Anderson Cancer Center)		
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	Exposure of crewmembers to space-related radiation can be a serious impediment to future space travel of long duration. There is a need to understand the effects of radiation on the immune system and the mechanism(s) of radiation-induced cell damage. In particular, we will study the effect of proton, heavy ion and gamma radiation on human dendritic cells (DCs) with respect to their ability to present antigen and its impact to the immune system. The period of performance for the work proposed is for 3 years, beginning June 1, 2002 through May 31, 2005. The recruitment of peripheral blood progenitor cells and the expansion of dendritic cells with hematopoietic growth factors has been used for reconstitution of the immune system. Dendritic cells are the most potent antigen presenting cells of the immune system and are required for optimal immune activation. Our proposal is to investigate the effects of radiation on monocyte-derived DC (MMDC) that will lead to a better understanding of the mechanism(s) involved and to propose the use of MDDC for immune restoration of crewmembers with immune suppression post space flight.		

Task Description:	Ground-based models have demonstrated adverse affects on the immune system due to stress and exposure to radiation. The biological effects of exposure to radiation in space may lead to serious health consequences for crewmembers of future space flights of long duration. The ability of immune cells such as MDDC and lymphocytes to survive exposure to radiation and the ability of the surviving cells to mediate an immune response is not well understood. During long-term space flight, astronauts may become immune compromised and develop secondary immunodeficiency and malignancies. The mechanism by which radiation exposure encountered during space flight of long duration may affect the immune system is not yet fully appreciated. Using the crewmember's own hematopoietic progenitor cells, lymphocytes, and MDDC that were collected prior to flight could be a countermeasure approach for reconstituting the immune system post radiation exposure. Even though much is known of the effects of radiation on different cell types, little is known of the maturation and function of MDDC exposed to radiation. We propose to study the effects of radiation, in particular, proton (available at the Loma Linda University), high-energy heavy ion beams (available at the Brookhaven Nat'l Lab.) and gamma-radiation on the differentialtion, maturation, and antigen presentation capability of human MDDC. Our hypothesis is that protracted space flight will reduce the number of dendritic cells, severely impair the immune system, and seriously attenuate immune surveillance mechanisms unless appropriate countermeasures are taken. The work will be performed at the University of Texas M. D. Anderson Cancer Center (MDACC) by Drs. James Reuben (Immunologist), Bang-Ning Lee (Molecular Biologist), and Martin Korbling, M.D. (Director of Apheresis Center, MDACC). The University of Texas M.D. Anderson Cancer Center ranks as one of the nation's most respected and productive Comprehensive Cancer Centers and is devoted exclusively to cancer patient care, research, education and prevention. We submit this proposal to study the effects of proton-, heavy ion- and gamma-radiation on monocyte-derived dendritic cells with the following specific aims: Aim #1: To determine the effect of proton-, heavy ion- and gamma-radiation on the differentiation of myeloid precursor cells into immature dendritic cells. Aim #2: To determine the effect of proton-, heavy ion- and gamma-radiation on the ability of immature DCs to capture antigen and to express costimulatory receptors. Aim #3: To determine the effect of proton-, heavy ion- and gamma-radiation on the ability of activated mature DCs to present antigen to T cells. Aim #4: To identify countermeasures that would obviate the adverse effects of proton-, heavy ion- and gamma-radiation on myelosuppression and the subsequent immune suppression. To determine the effects of low-dose radiation on the differentiation and functon of antigen presenting cells, dendritioc cells.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The work done on the effects of radiation is important as in clinical practice, many patients receive therapeutic radiation at much higher does than used in this study. We used doses no more than 15 Gy to study the effects on the human immune system. In particular, we studied the effects of low-dose radiation on T-cells and the antigen presenting cells, dendritic cells. This work will also help in the design of dendritic cell based vaccines that are becoming more in use for the treatment of cancer.
Task Progress:	Ground-based models have demonstrated adverse affects on the health of astronauts due to stress from zero gravity and exposure to radiation, such as high-energy ionizing radiation (HZE) on protracted space flight to Mars. Space or solar radiation is acknowledged to comprise photons (X-rays), electrons (gamma-rays), protons, neutrons, and heavy metal ions. Even though the dose rate of HZE in space would be low, little is known of the bystander effect of HZE even at low dose rates. In a series of experiments conducted at Loma Linda University (LLU), investigators carefully delineated the effects of proton radiation upon immune structure and function in murine models. C57BL/6 mice were given 3 Gray (Gy) protons in place of gamma rays in one dose; some animals also received an immunization with sheep red blood cells (SRBC). By days 4 to 10 after irradiation, statistically significant decreases were found in CD19+ B cells, CD3+ T cells, CD4+ T cells, and CD8+ T cells. Irradiated animals showed a delayed and lowered anti-SRBC antibody response. Natural killer cells were relatively radioresistant. B cells recovered by day 15 and CD3+ and CD4+ T cells by day 29, but CD8+ T cells remained impaired. These authors also discovered that the conventional gamma radiation yielded similar data. In a paired series of experiments in this murine model system, the LLU investigators found that gamma-ray treatment acutely (4 days) decreased spleen lymphocytes more than it did peripheral blood lymphocytes (PBL); higher doses were more toxic and acutely (4 days) decreased IL-2 secretion by activated splenocytes. These experiments provide ample evidence that space radiation is likely, at least acutely, to affect immune forces. Our previous study on gamma radiation demonstrated that g-irradiation of peripheral blood monocytes resulted in down-regulation of co-stimulatory molecules on monocyte-derived dendritic cells (MDDC). Others further demonstrated that gamma irradiation reduced expression of CD86, CD80 and HLA-DR molecules on human MDDC, especially CD86. Furthermore, irradiated DC showed a markedly lower capacity to promote T-cell proliferation, and strikingly up-regulated secretion of IL-4 and IL-2 by autologous T cells when compared to non-irradiated DC. In animal study, the shifted response of the helper T cells after whole body g irradiation is not only due to a significant suppression of the secretion of IL-12p70 in the antigen-presenting cells, but also to the lower expression of the IL-12 receptor on T cells. The initial immunological events in response to infection include antigen uptake by professional antigen-presenting cells (APC), processing of antigen into immunogenic peptides, and presentation in the context of major histocompatibility complex (MHC) molecules to naïve T cells. The two main subsets of APC, myeloid dendritic cells (MDC) and plasmacytoid dendritic cells (PDC), can be distinguished from one another based on mutually exclusive expression of leukocyte integrin (CD11c) or the interleukin (IL)-3 receptor (CD123), respectively. In addition, PDC and MDC express a family of pattern recognition receptors, toll-like receptors (TLR) that are activated by specific components of microbes and certain host molecules. TLR constitute the first line of defense against many pathogens and play a crucial role in the function of the innate and adaptive immune system. Activation of the TLR-7 on PDC and TLR-8 on MDC leads to the secretion of interferon-alpha (IFN-a) and tumor necrosis factor-alpha (TNF-a) by PDC and interleukin-12 (IL-12) and TNF-a by MDC, respectively. IFN-a is a potent inhibitor of virus replication and potent inducer of innate immunity whereas TNF-a and IL-12 are potent inducers of adaptive immunity. In this year, we investigated the effect of low-dose g-irradiation on the ability of circulating PDC

and MDC activated through TLR-7 and TLR-8 with a synthetic ligand, CL097 to produce the prominent cytokines IFN-a, TNF-a, and IL-12. Our hypothesis is that low-dose g-irradiation of PDC and MDC would suppress their ability to secrete the aforementioned modulators of innate and adaptive immunity and interfere with the ability of the host to be protected from infectious agents.

We intend to investigate the effect of low dose gamma (g)-irradiation on the circulating dendritic cells DC, peripheral blood were irradiated with gamma-ray to determine the effect of low-dose g-irradiation on the ability of circulating plasmacytoid DC (PDC) and Myeloid DC (MDC) activated through TLR-7 and TLR-8 with a synthetic ligand, CL097 to produce the prominent cytokines IFN-a, TNF-a, and IL-12. The percentage and the mean channel of fluorescence intensity (MFI) of cytokine producing MDC and PDC were calculated. Non-parametric Friedman test was used to analyze the statistical differences of cytokine production among radiation doses.

Recently, we have accomplished the following based on the data obtained from 12 healthy normal subjects: 1. Compared with un-irradiated PDC or MDC, there were no significant differences in the percentage of cytokine-producing PDC or MDC subject to low-dose g-radiation. 2. There is an irradiation dose dependent reduction of the mean channel fluorescence intensity (MFI) of intracellular IFN-a (P = 0.004) and TNF-a (P = 0.000) of the plasmacytoid DC upon stimulation with 5 uM of CL097. 3. The amount of IL-12 (p = 0.007) and TNF-a (p = 0.028) produced by individual MDC was significantly reduced upon the activation through TLR-8 with CL097 in an irradiation dose-dependent manner.

In summary, the suppression of PDC to synthesize IFN-a and TNF-a by low-dose g-irradiation is unprecedented and suggests the possible attenuation of innate immune surveillance, whereas the reduced production of IL-12 and TNF-a by low-dose g-irradiated MDC might negatively impact the ability of MDC to mediate adaptive immunity and effector T-cell function. As DC are essential for the development of an immune response and clearance of microbial infection, low-dose g-radiation might pose a serious threat to long-term health in space.

Bibliography Type:	Description: (Last Updated: 08/05/2020)
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